

Power Supply Noise in Analog Audio Class D Amplifiers

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Abstract—A potential drawback of class D amplifiers (CDAs) is their relatively poor tolerance to power supply noise. It has been recently established by means of a linear model that the manifested nonlinearities due to the supply noise are the fundamental noise frequency component and second-order intermodulation components, qualified by the power supply rejection ratio (PSRR) and intermodulation distortion (IMD), respectively. In this paper, multi-dimensional Fourier series analysis is applied to open-loop, single-, and double-feedback pulsewidth modulation CDAs, and expressions for PSRR, IMD, and fold-back distortion (FBD) are derived. This analysis takes into account the nonlinear pulse modulation process (unaccounted for in the linear model), and all components of PSRR and IMD nonlinearities are hence modeled. It is shown that the first-harmonic PSRR and third-order IMD components, which are usually ignored in CDAs and linear amplifiers, are significant and that, under certain conditions, the third-order IMD components can be higher than the second-order IMD components. It is also shown that the loop gain should be high for high PSRR and low IMD. Furthermore, significant FBD can arise due to the intermodulation between the supply noise, the input signal, and the carrier, and interestingly, the FBD in closed-loop CDAs is more serious than that in open-loop CDAs. The analyses herein are verified by computer simulations and on the basis of measurements on a prototype CDA IC and on other hardware realizations.

Index Terms—Class D amplifiers (CDAs), intermodulation distortion (IMD) pub.

I. INTRODUCTION

AUDIO POWER amplifiers are increasingly based on a class D amplifier (CDA) architecture due to its substantially higher power-efficiency attribute over their classical linear analog class A and AB counterparts [1]. This attribute is largely due to their output-stage switching-mode operation and is particularly attractive for remotely powered devices and for their small form factor. The pulse modulation techniques often employed in analog CDAs include the following: 1) pulsewidth modulation (PWM) [2]; 2) pulse density modulation [3]; and 3) bang-bang modulation [4]; digital CDAs are often based on 1) and 2) [5], [6]. Among these approaches, the PWM approach is arguably the most prevalent due to its relatively lower switching frequency, high stability at near-100% modulation [7], and its simpler architecture, and is the approach of interest in this paper.

Power supply noise, often qualified by power supply rejection ratio (PSRR), and its mechanisms are well established for

linear amplifiers [8]. The supply noise has been recently recognized [9] as a drawback of CDAs compared with their linear counterparts and may otherwise require expensive hardware solutions such as highly stabilized low-noise power supplies. This drawback arises primarily because of the architecture of the switching-mode output stage. For example, the fundamental PSRR of an open-loop CDA is merely 6 dB [10], [11]; in this paper, the fundamental PSRR refers to the PSRR due to the nonlinear component whose frequency is the same frequency as the frequency of supply noise, typically 100 Hz or 120 Hz. On the basis of a linear model [11] in the Laplace domain, we have recently modeled the mechanisms and derived analytical expressions for the fundamental PSRR of the following three prevalent CDA design topologies: 1) open-loop CDA; 2) single-feedback (1FB-) CDA; and 3) double-feedback (2FB-) CDA. Topologies 2) and 3) are closed-loop CDAs.

In addition to the supply noise at the CDA output qualified by PSRR, we have recently shown [11] that the supply noise may further intermodulate with the input signal at the output stage and manifest into intermodulation distortion (IMD). In this case, the IMD is the second-order IMD wherein the frequencies of the nonlinear components are equal to the input signal frequency plus/minus the supply noise frequency; note the intermodulation signals herein and the definition of IMD as opposed to the usual IMD due to the intermodulation between two input signals. The manifested IMD, although not well recognized within the electronics community, can be serious. For example, the second-order IMD of an open-loop CDA is -9 dB for the worst-case condition where the modulation index $M = 1$ and with respect to the supply noise.

Although the earlier derived linear model [11] for PSRR and IMD is intuitive and useful, as it provides insight into the mechanisms and the pertinent parameters therein, the linear model is incomplete in the sense that it does not model all nonlinearities due to the supply noise, including the harmonic PSRR components and the higher order IMD components. This shortcoming is primarily because the PWM operation/process was assumed to be linear when it is actually nonlinear (as modeled by Fourier series). In this paper, the harmonic PSRR refers to the PSRR whose components are at multiple integers of the supply noise frequency, and the higher order IMD refers to the IMD whose components are at frequencies that are the input signal frequency plus/minus the multiple integers of the supply noise frequency. The additional nonlinear components, namely, the harmonic PSRR and higher order IMD, can be significant in densely populated system-on-chip environments where the supply noise may include high-frequency components (as opposed to only the fundamental low-frequency noise) due to the

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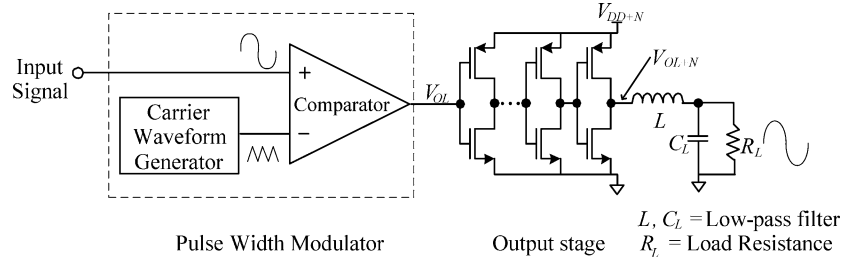


Fig. 1. Schematic of an open-loop CDA.

presence of low- and high-frequency mixed-signal, radio-frequency, and switching power supply circuits. In short, harmonic PSRR and higher order IMD of CDAs remain unreported in literature.

Another well-known nonlinearity in CDAs is fold-back distortion (FBD) [5], [12], and conventionally, FBD is due to the intermodulation between the input signal and the carrier. However, it is not well known and is unreported in literature that another FBD arises when the supply noise intermodulates with both the carrier and the input signal. In this paper, the FBD comprises both the conventionally defined FBD and the manifested FBD (defined in Section III-D) due to the intermodulation of these three signals. FBD was not accounted for in the earlier linear model, and this is because of the same assumption on the PWM process wherein all frequency components related to the carrier are ignored.

It is generally agreed within the electronics community that nonlinearities can be effectively suppressed by negative feedback. This remains true for most nonlinearities in CDAs, including PSRR and IMD. However, in the case of FBD, we will later show that the converse is true and that, in some designs, the FBD can be serious (see Section IV later). For example, the FBD of a closed-loop CDA is unexpectedly higher than that of the open-loop CDA at full modulation ($M = 1$) and carrier frequency $f_c = 80$ kHz; a low carrier frequency is often required in power-critical applications such as hearing aids.

In this paper, we derive multidimensional Fourier series expressions to analytically determine all nonlinear components of the PSRR, IMD, and FBD nonlinearities for the prevalent open-loop, 1FB-, and 2FB-CDA topologies, and thereafter derive analytical expressions to determine PSRR, IMD, and FBD. The analyses provide insight into the mechanisms and parameters of these nonlinearities and how the various pertinent parameters may be varied/compromised to meet a given set of specifications. The derived expressions are verified by comparing the numerical values obtained from the derived expressions (via MATLAB) against that obtained from HSPICE simulations, and on the basis of measurements on a prototype CDA IC and on other CDAs realized by using discrete components. It is worthwhile to remark that the derived expressions, based on the 1FB- and 2FB-CDAs, can be easily extended to infer the different nonlinear components of other closed-loop CDA topologies. For completeness, it is perhaps of interest to note that we have earlier derived a double Fourier series expression to analytically determine the total harmonic distortion of both open- and closed-loop CDAs [13], [14].

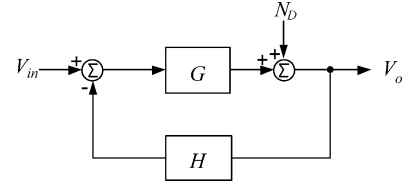


Fig. 2. Generic model of a closed-loop CDA.

This paper is organized in the following manner. In Section II, a succinct review of our earlier derived linear model is presented. In Section III, the Fourier series expression to determine all components of PSRR, IMD and FBD nonlinearities for open-loop, 1FB- and 2FB-CDAs, and thereafter analytical expressions to determine PSRR, IMD and FBD, are derived. In Section IV, the derived expressions are verified against HSPICE simulations and on the basis of practical measurements. Finally, conclusions are drawn in Section V.

II. REVIEW OF THE LINEAR MODEL FOR PSRR AND IMD

This section succinctly reviews the prevailing linear model [11] based on the Laplace approach, and serves as a preamble to the latter derivations based on the multidimensional Fourier series approach.

Fig. 1 shows the open-loop CDA [13] where V_{DD+N} is the supply voltage plus noise. We earlier [11] expressed the noise components N_D at V_{DD+N} as

$$N_D(t) = \frac{N}{2} \cos \omega_n t + \frac{MN}{4} \cos(\omega_s t \pm \omega_n t) \quad (1)$$

where M is the modulation index, ω_s is the angular frequency of the input signal, N is the normalized amplitude of the supply noise with respect to V_{DD} , and ω_n is the angular frequency of the supply noise.

For the sake of simplicity of analysis and without loss of generality, the noise components expressed in (1) are normalized with respect to the power supply V_{DD} . Equation (1) comprises two terms. The first term is the fundamental PSRR component, and the second term comprises the second-order IMD components. It has been shown [11] that the fundamental PSRR and second-order IMD of the open-loop CDA are 6 dB and approximately -9 dB in the worst case when $M = 1$, respectively. Of specific interest, it was shown that no design parameters are available to the designer to improve the PSRR and IMD.

Fig. 2 shows a generic linear model of a closed-loop CDA (e.g., the design in Fig. 3), where G is the forward loop gain and

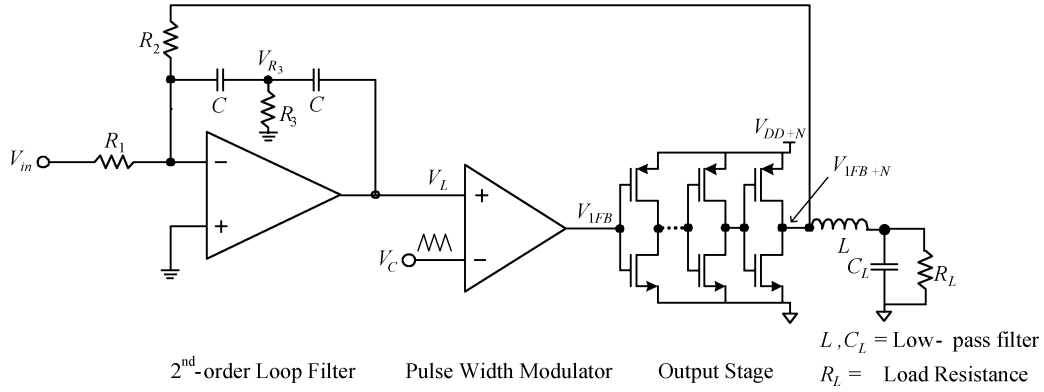


Fig. 3. Schematic of a 1FB-CDA.

H is the feedback factor. This linear model [11] is essentially the well-known linear control approach based on Laplace analysis, wherein the pulsewidth modulator is assumed to be a linear stage with a single-frequency component and an effective gain.

Based on the linear model [11], the fundamental PSRR and second-order IMD of the closed-loop CDA are shown in (2a) and (2b), at the bottom of the page, respectively, where $LG(\omega)$ is the loop gain GH at ω .

This model shows that the PSRR and IMD can, as expected, be improved by increasing the loop gain. Practically, this loop gain is the product of the forward loop gain and the feedback factor, where gains of the loop filter and the pulsewidth modulator (see Fig. 3) are parameters of the forward loop gain. Note that this model considers only the fundamental PSRR component (ω_n) and second-order IMD components ($\omega_s \pm \omega_n$).

In the next section, the multidimensional Fourier series analysis approach which accounts for the harmonic PSRR, the higher order IMD, and the FBD nonlinear components (all due to the supply noise) will be adopted for the open-loop, 1FB-, and 2FB-CDAs. For completeness, note that some researchers deem the multidimensional Fourier series analysis a “nonlinear” analysis because it takes into consideration all frequency components (of which some are absent in the linear model) in the output PWM signal.

III. FOURIER SERIES ANALYSIS ON PSRR, IMD, AND FBD

In Section III-A, the Fourier series analysis is first applied to the open-loop CDA with a noisy supply to obtain a complete expression for its PWM output. Following this, in Section III-B, the Fourier series analysis is applied to the 1FB-CDA. In Section III-B1, the Fourier series expression for the output PWM signal of an ideal 1FB-CDA with a noise-free supply is derived, and the derived expression is thereafter employed to derive expressions for the PSRR and IMD for the 1FB-CDA

with a noisy supply in Section III-B2. In Section III-C, the derived PSRR and IMD expressions in Section III-B2 are extended to the 2FB-CDA. In Section III-D, on the basis of the Fourier series expressions for the output PWM signal in the open- and closed-loop CDAs with noisy supply derived in Sections III-A and III-B2, respectively, Fourier series analysis is finally applied to derive expressions for FBD for open- and closed-loop CDAs.

A. Fourier Series Analysis of the Open-Loop CDA

In this section, the double Fourier series expression for the open-loop CDA will be derived, and this serves as a preamble to the derivation of the expressions for closed-loop CDAs.

The established expression of the output PWM signal of an ideal open-loop CDA [15] is

$$\begin{aligned}
 V_{OL}(t) = & 0.5 + \frac{M}{2} \cos \omega_s t \\
 & + 2 \sum_{m=1}^{\infty} \left(\frac{J_0\left(m\pi \frac{M}{2}\right)}{m\pi} \sin m\omega_c t \cos \frac{m\pi}{2} \right) \\
 & + 2 \sum_{m=1}^{\infty} \sum_{n=\pm 1}^{\pm \infty} \left(\frac{J_n\left(m\pi \frac{M}{2}\right)}{m\pi} \right. \\
 & \quad \times \cos(m\omega_c t + n\omega_s t) \\
 & \quad \times \sin\left(\frac{m\pi + n\pi}{2}\right) \left. \right) \quad (3)
 \end{aligned}$$

where ω_c is the angular carrier frequency and $J_n()$ is the n th-order Bessel function of the first kind.

The last transistors in the output stage in Fig. 1 may be modeled as an on-resistance with an ideal on/off switch. Due to the voltage drop across the on-resistance of the output pMOS transistor when it is switched on, the top envelope voltage of the output PWM signal is slightly smaller than the supply voltage.

$$PSRR_{CL} = 20 \log [2 + 2LG(\omega_n)] \quad (2a)$$

$$IMD_{CL} = 20 \log \left(\frac{M}{4} \times \sqrt{[1 + LG(\omega_s + \omega_n)]^{-2} + [1 + LG(\omega_s - \omega_n)]^{-2}} \right) \quad (2b)$$

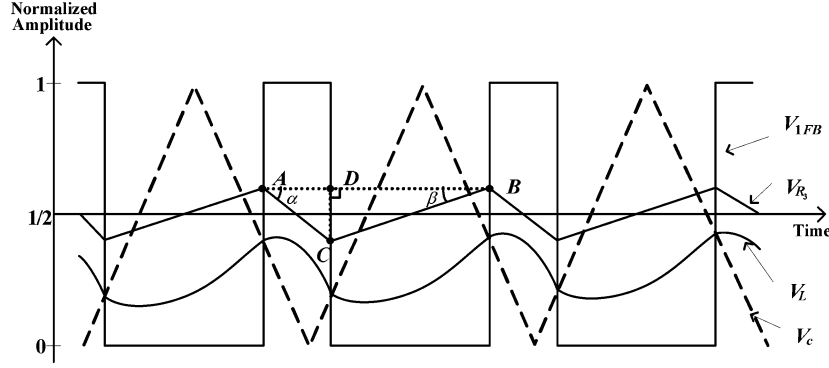


Fig. 4. Waveforms of V_{R3} , and two inputs (V_L , V_c) and one output (V_{1FB}) of the comparator.

In addition to the supply noise at the fundamental frequency appearing at the CDA output at V_{OL+N} in Fig. 1, supply noise harmonics are also produced due to the varying on-resistance of the output stage transistor modulated by the supply noise. However, as we can show that these noise harmonics are insignificant compared to the fundamental frequency supply noise, the actual noisy supply can be used as an approximation of the top envelope voltage of the output PWM signal in the open-loop CDA and, later, in the closed-loop CDA.

On the basis of (3), the output PWM signal of the open-loop CDA with nonlinearities due to the supply noise can easily be derived as follows:

$$\begin{aligned}
 V_{OL+N}(t) &= V_{OL}(t) \times (1 + N \cos \omega_n t) \\
 &= \frac{1}{2} + \frac{M}{2} \cos \omega_s t + \frac{N}{2} \cos \omega_n t + \frac{MN}{4} \cos(\omega_s t \pm \omega_n t) \\
 &\quad + 2 \sum_{m=1}^{\infty} \left(\frac{J_0\left(\frac{m\pi}{2}\right)}{m\pi} \sin m\omega_c t \cos \frac{m\pi}{2} \right) \\
 &\quad + \sum_{m=1}^{\infty} \left(\frac{J_0\left(\frac{m\pi}{2}\right)}{m\pi} N \cos(m\omega_c t \pm \omega_n t) \sin \frac{m\pi}{2} \right) \\
 &\quad + \sum_{m=1}^{\infty} \sum_{n=\pm 1}^{\pm \infty} 2J_{OL} \cos(m\omega_c t + n\omega_s t) \\
 &\quad + \sum_{m=1}^{\infty} \sum_{n=\pm 1}^{\pm \infty} NJ_{OL} \cos(m\omega_c t + n\omega_s t \pm \omega_n t) \quad (4)
 \end{aligned}$$

where

$$J_{OL} = \frac{J_n\left(\frac{m\pi}{2}\right)}{m\pi} \sin\left(\frac{m\pi + n\pi}{2}\right).$$

Equation (4) is interpreted as follows. The first two terms are an inconsequential dc offset and the amplified replica of the input signal, respectively. The third and fourth terms are the same as that given in (1). The nonlinear components of the fundamental PSRR and second-order IMD are represented by the third and fourth terms, respectively, and were already shown to be poor, which are PSRR ≈ 6 dB at ω_n and IMD ≈ -9 dB at $M = 1$ and at $\omega_s \pm \omega_n$, respectively. Note that because the CDA is open loop, the harmonic PSRR components and third-order or

higher order IMD components are absent. The fifth term comprises the carrier and its harmonics and is inconsequential because they are beyond the audio range. The sixth term comprises the intermodulation between the carrier and the supply noise and is insignificant (lower than the floor noise of ~ -95 dB). The seventh term comprises the intermodulation between the input signal and the carrier. The last term comprises the intermodulation between the following three signals—the input signal, carrier, and supply noise. The last two terms contain the FBD components and will be discussed in Section III-D later.

On the basis of the derivation of the double Fourier series expression for the PWM output of the open-loop CDA with a noisy supply, the expressions for PSRR and IMD for closed-loop CDAs will now be derived.

B. Fourier Series Analysis of the 1FB-CDA

In this section, a Fourier series expression for the 1FB-CDA will be derived, and thereafter, the expressions of PSRR and IMD are derived and interpreted.

1) *Fourier Series Expression for Noise-Free Supply and Infinite Loop Gain:* Fig. 3 shows a 1FB-CDA comprising a second-order loop filter (integrator), a pulsewidth modulator, an output stage, and the load comprising an LC low-pass filter and a load (R_L). At the outset, note that V_{1FB} and V_{1FB+N} are both out of phase with respect to the input signal. However, to be consistent with the previous derivation for the open-loop CDA and for convenience of interpretation, V_{1FB} and V_{1FB+N} will henceforth be assumed to be in phase with the input signal. There is no loss of generality in making this assumption.

The derivation of the expressions for the PWM signal, V_{1FB} , involves two steps. In step 1, the duty cycle of the PWM signal is derived, and in step 2, V_{1FB} is obtained by applying the derived duty cycle to a general Fourier series expression for rectangular pulses. The ideal case of a noise-free supply and infinite loop gain is assumed here, and the practical case, where the supply is noisy and the loop gain is finite, will be considered in the following section.

Step 1—Derivation of the Duty Cycle $\zeta(t)$: Fig. 4 shows the time-domain waveforms of the voltage across R_3 (V_{R3}), the output of loop filter (V_L), the triangular carrier signal (V_c), and the PWM signal at the comparator output (V_{1FB}). To simplify the derivation, V_c is assumed to be full swing (i.e., the gain of the pulsewidth modulator is 1), and this assumption does not affect

the final output PWM expression. Although it is obvious that the duty cycle of V_{1FB} is determined by the comparisons between V_L and V_c , it is simpler to derive the duty cycle of V_{1FB} based on the first-order integrated signal at V_{R3} , i.e.,

$$V_{R3}(t) = -\frac{1}{\tau_1} \int V_{in} dt - \frac{1}{\tau_2} \int \left(V_{1FB} - \frac{1}{2} \right) dt \quad (5)$$

where $\tau_1 = R_1 C$ and $\tau_2 = R_2 C$ are the time constants of the integrator.

Assuming that V_{in} is constant within one period, ΔABC in Fig. 4 can be drawn, wherein A , C , and B are the intersection points between V_{R3} and V_{1FB} , and line CD is drawn perpendicular to line AB . Line AB represents one period of V_{1FB} , and line AD represents the time interval for which V_{1FB} is “1.” From (5), the slopes of lines AC and CB when V_{1FB} is “1” and “0” are, respectively

$$\tan \alpha = \frac{V_{in}}{\tau_1} + \frac{1/2}{\tau_2} \quad \tan \beta = \frac{1/2}{\tau_2} - \frac{V_{in}}{\tau_1}. \quad (6)$$

The time intervals represented by lines DB and AD can then be expressed as

$$DB = \frac{\tan \alpha}{\tan \alpha + \tan \beta} d \quad AD = \frac{\tan \beta}{\tan \alpha + \tan \beta}. \quad (7)$$

Assuming that V_{1FB} is in phase with the input signal, the duty cycle $\zeta(t)$ can be derived as

$$\zeta(t) = \frac{DB}{AB} = \frac{1}{2} + \frac{\tau_2}{\tau_1} V_{in} = \frac{1 + M \cos \omega_s t}{2}. \quad (8)$$

In (8), note that the information signal represented by $\zeta(t)$ of the PWM signal is the input signal amplified by τ_2/τ_1 .

Step 2—Derivation of the Output PWM Signal V_{1FB} : The general Fourier series expression of rectangular pulses is given by

$$\begin{aligned} V(t) &= V_0(t) + \sum_{m=1}^{\infty} V_m(t) \\ &= \zeta(t) + \sum_{m=1}^{\infty} \frac{2}{m\pi} \sin m\pi \zeta(t) \cos m\omega_c t. \end{aligned} \quad (9)$$

Substituting (8) into (9), the PWM signal expression is

$$\begin{aligned} V_{1FB}(t) &= \frac{1}{2} + \frac{M \cos \omega_s t}{2} \\ &+ \sum_{m=1}^{\infty} \sum_{q=-\infty}^{\infty} \frac{1}{m\pi} [B_o \sin(m\omega_c - q\omega_s)t \\ &+ B_e \cos(m\omega_c - q\omega_s)t] \end{aligned} \quad (10)$$

where

$$\begin{aligned} B_o &= \frac{1 - (-1)^q}{2} J_q \left(m \frac{M\pi}{2} \right) \cos \frac{(m+q)\pi}{2} \\ B_e &= \frac{1 + (-1)^q}{2} J_q \left(m \frac{M\pi}{2} \right) \sin \frac{(m+q)\pi}{2}. \end{aligned}$$

Equation (10) is interpreted as follows. The first two terms are, as in (4) for the open-loop CDA, the inconsequential constant dc offset and the amplified replica of the input signal, respectively. The third term comprises the components related to the carrier,

which are the carrier, its harmonics, and the intermodulation components between the input signal and the carrier. Although (10) is based on the 1FB-CDA, we will later show in Section III-C that the derived Fourier series expression can be easily extended to other closed-loop topologies, e.g., the 2FB-CDA.

In summary, the double Fourier series expression for the PWM signal of the 1FB-CDA where the supply is noise free and the loop gain is assumed to be ideal (∞) has been derived.

2) *PSRR and IMD in the 1FB-CDA:* Consider now a practical case where the power supply is noisy and the loop gain is finite. The same procedure in the preceding section will be applied.

In step 1, V_{1FB+N} is fed back through the feedback path formed by R_2 . To account for the supply noise, the slopes of lines AC and CB are rederived by adding the supply noise into the integration in (5) when V_{1FB} is “1.” Furthermore, to account for the effect of finite loop gain in a practical realization with supply noise, the error term V_e (see (14c) later) is added into the derivation to modify the duty cycle of the CDA with infinite loop gain. The slopes of lines AC and CB are rederived as

$$\begin{aligned} \tan \alpha &= \frac{V_{in}}{\tau_1} + \frac{V_n + 1/2 - V_e}{\tau_2} \\ \tan \beta &= \frac{1/2 + V_e}{\tau_2} - \frac{V_{in}}{\tau_1} \end{aligned} \quad (11)$$

where V_n is the supply noise in V_{DD+N} .

Substituting (11) into (7) and (8), the duty cycle $\zeta_n(t)$ of the PWM signal with power supply noise at V_{1FB} is

$$\zeta_n(t) = \frac{1/2 + M/2 \cos \omega_s t + N \cos \omega_n t - V_e}{1 + N \cos \omega_n t}. \quad (12)$$

In step 2, substituting (12) into (9), the Fourier series expression for the PWM signal at the comparator output is

$$V_{1FB}(t) = \zeta_n(t) + \sum_{m=1}^{\infty} \frac{2}{m\pi} \sin m\pi \zeta_n(t) \cos m\omega_c t. \quad (13)$$

To account for the supply noise injected at the output stage, the supply noise is now augmented into (13). Assuming that the output PWM signal V_{1FB+N} is in phase with the input signal, the final expression of output PWM signal, taking into consideration the supply noise and finite loop gain, is (refer to the Appendix)

$$V_{1FB+N}(t) = 1 - [1 - V_{1FB}(t)](1 + N \cos \omega_n t) \quad (14a)$$

$$= \frac{1}{2} + \frac{M \cos \omega_s t}{2} + V_{1FB+N}(t)_3 - V_e \quad (14b)$$

where

$$\begin{aligned} V_{1FB+N}(t)_3 &= \sum_{m=1}^{\infty} \sum_{a=-\infty}^{\infty} \sum_{b=-\infty}^{\infty} \sum_{c=-\infty}^{\infty} \sum_{d=-\infty}^{\infty} J_{CL} \\ &\times \{ B_1 N \cos(k_3 \pi) \sin[k_1 \omega_s + (k_2 + 1)\omega_n + m\omega_c]t \\ &+ B_1 N \cos(k_3 \pi) \sin[k_1 \omega_s + (k_2 - 1)\omega_n + m\omega_c]t \\ &+ B_2 N \sin(k_3 \pi) \cos[k_1 \omega_s + (k_2 - 1)\omega_n + m\omega_c]t \\ &+ B_2 N \sin(k_3 \pi) \cos[k_1 \omega_s + (k_2 + 1)\omega_n + m\omega_c]t \\ &+ 2B_1 \cos(k_3 \pi) \sin(k_1 \omega_s + k_2 \omega_n + m\omega_c)t \\ &+ 2B_2 \sin(k_3 \pi) \cos(k_1 \omega_s + k_2 \omega_n + m\omega_c)t \} \end{aligned} \quad (14c)$$

$$B_1 = [1 - (-1)^{k_1+k_2}]/4 \quad B_2 = [1 + (-1)^{k_1+k_2}]/4$$

$$k_1 = -a + c + d \quad k_2 = -b - c + d$$

$$k_3 = (a + b + m - c - d)/2$$

$$J_{CL} = \frac{1}{m\pi} J_a \left(\frac{mM\pi}{2} \right) J_b \left(\frac{mN\pi}{2} \right) \times J_c \left(\frac{mMN\pi}{4} \right) J_d \left(\frac{mMN\pi}{4} \right)$$

$$V_e \approx \frac{N \cos \omega_n t}{2(LG(\omega_n) + 1)} + \frac{MN \cos(\omega_s \pm \omega_n)t}{4(LG(\omega_s \pm \omega_n) + 1)} + \frac{N^2 \cos 2\omega_n t}{4(LG(2\omega_n) + 1)} + \frac{MN^2 \cos(\omega_s \pm 2\omega_n)t}{8(LG(\omega_s \pm 2\omega_n) + 1)} \quad (14d)$$

with

$$LG = H_1 \times G_{int1} \times G_{PWM} \quad (14e)$$

$H_1 = R_1/(R_1 + R_2)$ being the feedback factor, G_{int1} being the loop filter gain, and $G_{PWM} = V_{DD}/V_c$.

Equation (14b) is interpreted as follows. As in the previous pertinent equations, the first and second terms comprise the inconsequential dc offset and the amplified input signal. The third term (14c) is the intermodulation between the input signal, the supply noise, and the carrier and further includes the FBD components that will be presented in Section III-D later. The last term (14d) is the error term due to the supply noise and finite loop gain and comprises the fundamental and first-harmonic PSRR components and second- and third-order IMD components; as described in the Appendix, only the significant nonlinear components are considered in (14d).

Equation (14d) is interpreted as follows. The first term is the fundamental PSRR component at ω_n , arising directly from the supply noise. The second term comprises the second-order IMD components at $\omega_s \pm \omega_n$. The third term is the first-harmonic PSRR component at $2\omega_n$. The fourth term comprises the third-order IMD components at $\omega_s \pm 2\omega_n$. To put (14d) in perspective, if the third and fourth terms are ignored, the PSRR and IMD components therein are identical to that in (2a) and (2b) based on the linear model.

As verification, the time-domain waveforms obtained from (14b) using MATLAB agree well with the HSPICE simulations of the circuit in Fig. 3.

Finally, on the basis of V_e (14d), the PSRR and IMD of the 1FB-CDA are

$$PSRR \approx PSRR(\omega_n) + PSRR(2\omega_n) \quad (15a)$$

where

$$PSRR(\omega_n) = 20 \log(2LG(\omega_n) + 2) \text{ dB} \quad (15b)$$

$$PSRR(2\omega_n) \approx 20 \log \left(\frac{4LG(2\omega_n) + 4}{N} \right) \text{ dB} \quad (15c)$$

and

$$IMD \approx IMD(\omega_s \pm \omega_n) + IMD(\omega_s \pm 2\omega_n) \quad (16a)$$

where $IMD(\omega_s \pm \omega_n)$ and $IMD(\omega_s \pm 2\omega_n)$ are shown at the bottom of the page.

As delineated earlier, only the fundamental and first-harmonic PSRR components are considered in (15), as the remaining higher harmonics are negligible. Similarly, in (16), only the second- and third-order IMD components are considered, as the remaining higher order IMD components are negligible. From (15) and (16), a practical observation can be made—a higher loop gain increases the PSRR and reduces the IMD. Furthermore, as expected, note that the IMD increases as the magnitude of input signal increases.

The significance of PSRR and IMD expressed by (15a) and (16a) over that for the linear CDA model is substantial in some practical designs. Specifically, this is unlike what is generally observed in linear amplifiers where the first- (and higher) harmonic PSRR components (15c) and third-order (and higher order) IMD components (16c) are insignificant compared to their fundamental (15b) and second-order (16b) counterparts, respectively—this is not necessarily the case in CDAs. The significance of these first-harmonic PSRR and third-order IMD components in the 1FB-CDA will now be described.

To compare the first-harmonic PSRR against its fundamental, the ratio of the nonlinearity magnitudes at $2\omega_n$ and ω_n is taken, which is equivalent to the difference of the corresponding PSRRs

$$PSRR(2\omega_n) - PSRR(\omega_n) \approx 20 \log \left(\frac{2[LG(2\omega_n) + 1]}{N[LG(\omega_n) + 1]} \right) \text{ dB}. \quad (17)$$

From (17), there are two parameters that determine the significance of the first-harmonic PSRR. First, the parameter N is usually small (< 0.1), and the first-harmonic PSRR is hence usually insignificant. Nonetheless, in some noisy systems, N can be large, and the effect of the first-harmonic component increases. Second, as $LG(2\omega_n)$ can be substantially smaller than $LG(\omega_n)$ due to rapid decay of LG at relatively high frequencies, the first-harmonic PSRR can be significant. For example, in practice, it is possible that $LG(\omega_n) > 4LG(2\omega_n)$, and in the example in Section IV, the first harmonic contributes $\sim 30\%$ of the total PSRR noise.

$$IMD(\omega_s \pm \omega_n) = 20 \log \left(\frac{M}{4} \times \sqrt{[LG(\omega_s + \omega_n) + 1]^{-2} + [LG(\omega_s - \omega_n) + 1]^{-2}} \right) \text{ dB} \quad (16b)$$

$$IMD(\omega_s \pm 2\omega_n) \approx 20 \log \left(\frac{MN}{8} \times \sqrt{[LG(\omega_s + 2\omega_n) + 1]^{-2} + [LG(\omega_s - 2\omega_n) + 1]^{-2}} \right) \text{ dB} \quad (16c)$$

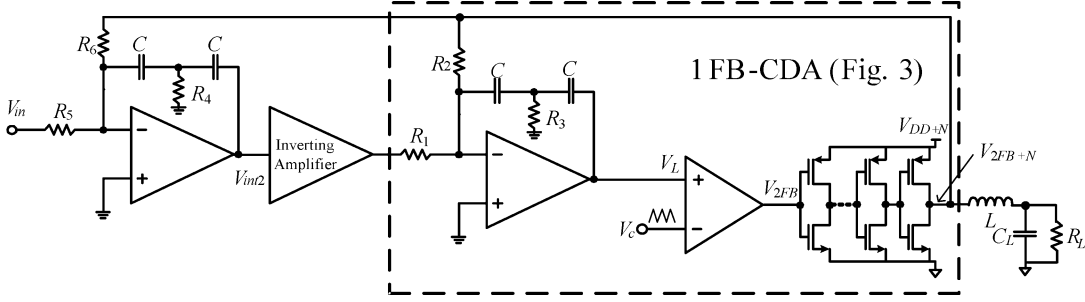


Fig. 5. Schematic of a 2FB-CDA.

Similarly, the difference between the third- and second-order IMDs is expressed as (18), shown at the bottom of the page.

From (18), note that the same two parameters, i.e., N and LG , determine the significance of the third-order IMD with respect to the second-order IMD. The effect of N on IMD is similar to that on PSRR. However, the effect of LG on IMD is different. In the case where both the input signal and the supply noise are at high frequencies, the IMD components at $\omega_s \omega_n$ and $\omega_s + 2\omega_n$ are inconsequential, as they are beyond the frequencies of interest. However, the remaining second-order IMD component at $\omega_s - \omega_n$ is low because $LG(\omega_s - \omega_n)$ is rather large, and the remaining third-order IMD component at $\omega_s - 2\omega_n$ is relatively larger because $LG(\omega_s - 2\omega_n)$ is greatly degraded at high frequencies. As a result, the third-order IMD can be even higher than the second-order IMD. For example, it is possible that $LG(\omega_s - \omega_n) > 40LG(\omega_s - 2\omega_n)$, and in the example in Section IV, the third-order IMD is more significant (10 dB worse) than the second-order IMD. This observation is particularly significant because, in linear amplifiers, the third-order IMD components are usually negligible.

From the PSRR and IMD expressions in (15) and (16), the magnitude of these nonlinearities can be reduced by a higher loop gain and is independent of the carrier frequency. However, in the practical design of closed-loop CDAs, the magnitude of the loop gain is usually constrained by the carrier frequency for maximum signal swing considerations, hence the signal-to-noise ratio. As we have delineated elsewhere [11], if a higher carrier frequency is tolerated (at the cost of lower power efficiency), a higher loop gain can be obtained because the magnitude of the carrier signal can be attenuated (by the loop filter) to the same degree due to the higher carrier frequency. This results in the desirable higher PSRR and lower IMD.

In summary, with Fourier series analyses, all significant components of PSRR and IMD have been modeled, and expressions of PSRR and IMD have been derived. Furthermore, unlike linear amplifiers, the first-harmonic PSRR components can be significant, and the third-order IMD components can be even higher than the second-order IMD components.

C. Fourier Series Analysis of the 2FB-CDA

Fig. 5 shows a 2FB-CDA, comprising a 1FB-CDA, input and feedback resistors, a second-order loop filter, an inverting amplifier, and a load. The inverting amplifier ensures an overall negative feedback and can be eliminated if the 1FB-CDA is non-inverting. The additional second-order loop filter and feedback path provide an enhanced loop gain for the 2FB-CDA over the 1FB-CDA. The loop gain of the 2FB-CDA can be expressed as

$$LG(2FB) = H_1 G_{int1} G_1 G_{int2} G_{PWM} + H_2 G_{int2} G_{PWM} \quad (19)$$

where G_{int1} is the loop gain of the inner second-order loop filter (of the 1FB-CDA), $G_1 = R_2/(R_2 + R_1)$, G_{int2} is the loop gain of the other second-order loop filter, and $H_2 = R_5/(R_5 + R_6)$ is the feedback factor.

Using the similar procedure delineated earlier in Section III-B, the duty cycle of the output PWM signal at V_{2FB+N} can be derived based on the first-order integrated signal at V_{int2} ; note that because of the typical values of C and R_4 in Fig. 5, the additional loop filter realizes a double-pole single-zero function, where the first pole and zero are close to each other. Furthermore, the expressions of the PSRR and IMD of the 2FB-CDA can also be expressed by (15) and (16), respectively, but with the associated loop gain of the 2FB-CDA (19).

As in the case of the PSRR and IMD of the 1FB-CDA, the PSRR of the 2FB-CDA comprises the significant fundamental and first-harmonic PSRR components, and the IMD of the 2FB-CDA comprises the significant second- and third-order IMD components. Also, as in the 1FB-CDA, the same practical observation on the loop gain can be made—a higher loop gain increases the PSRR and reduces the IMD.

When the 2FB-CDA is compared with the 1FB-CDA, the former is advantageous because the former's loop gain is higher than the latter, and hence its improved PSRR and IMD. It can

$$IMD(\omega_s \pm 2\omega_n) - IMD(\omega_s \pm \omega_n) \approx 20 \log \left(\frac{N}{2} \times \sqrt{\frac{[LG(\omega_s + 2\omega_n) + 1]^{-2} + [LG(\omega_s - 2\omega_n) + 1]^{-2}}{[LG(\omega_s + \omega_n) + 1]^{-2} + [LG(\omega_s - \omega_n) + 1]^{-2}}} \right) \text{ dB}. \quad (18)$$

also be shown that for the 2FB-CDA, the relationships between the fundamental and first-harmonic PSRR and between the second- and third-order IMDs are similar to that in the 1FB-CDA. Specifically, the first-harmonic PSRR component may be substantial (e.g., ~ 10 dB better than the fundamental components only), and the 3rd-order IMD components may be worse than the second-order IMD components. However, due to the substantially high loop gain of the 2FB-CDA, some PSRR and IMD components could be below the floor noise.

In summary, all significant components of PSRR and IMD have been modeled, and expressions for PSRR and IMD are derived, for the 2FB-CDA. For both the 1FB-CDA and the 2FB-CDA, their PSRR and IMD are largely determined by their respective loop gains. In this respect, the 2FB-CDA has worthy advantages over the 1FB-CDA due to its higher loop gain but at the cost of increased hardware.

D. Fourier Series Analysis of FBD in Open- and Closed-Loop CDAs

Consider now the FBD nonlinearity. The FBD herein comprises both the conventionally defined FBD (due to the intermodulation between the input signal and the carrier, abbreviated $\text{FBD}(V_{\text{in}} + V_c)$) and the now-defined FBD (that is due to the intermodulation between the supply noise, the input signal, and the carrier, abbreviated $\text{FBD}(V_{\text{in}} + V_c + V_n)$). Consider the open- and closed-loop CDAs in turn, and the FBD therein is quantified with respect to the modulation index.

In the open-loop CDA, from the last two terms of (4), the FBD can be expressed as

$$\text{FBD}_{\text{OL}} = \text{FBD}_{\text{OL}}(V_{\text{in}} + V_c) + \text{FBD}_{\text{OL}}(V_{\text{in}} + V_c + V_n) \quad (20)$$

where

$$\begin{aligned} & \text{FBD}_{\text{OL}}(V_{\text{in}} + V_c) \\ &= 20 \log \left(\frac{1}{M} \sum 2J_{\text{OL}}(m\omega_c + n\omega_s) \right) \text{ dB} \\ & \text{FBD}_{\text{OL}}(V_{\text{in}} + V_c + V_n) \\ &= 20 \log \left(\frac{1}{M} \sum N J_{\text{OL}}(m\omega_c + n\omega_s \pm \omega_n) \right) \text{ dB} \\ & 2\pi \times 20 \text{ Hz} < |n\omega_s + m\omega_c| < 2\pi \times 20 \text{ kHz} \\ & 2\pi \times 20 \text{ Hz} < |n\omega_s + m\omega_c \pm \omega_n| < 2\pi \times 20 \text{ kHz}. \end{aligned}$$

In (20), the first and last terms comprise the $\text{FBD}(V_{\text{in}} + V_c)$ and the $\text{FBD}(V_{\text{in}} + V_c + V_n)$ components, respectively. From (20), the three parameters that determine the FBD are as follows: 1) carrier frequency; 2) magnitude and frequency of the input signal; and 3) magnitude and frequency of the supply noise. We have previously delineated the effects of parameters 1) and 2) for $\text{FBD}(V_{\text{in}} + V_c)$ for both digital [5] and analog [16] CDAs. Specifically, the FBD is serious when the carrier frequency is low and the magnitude and frequency of the input signal are high. Parameter 3) is due to the intermodulation with the supply noise, and FBD is serious when the supply noise frequency is high. In general, the FBD due to $\text{FBD}(V_{\text{in}} + V_c + V_n)$ is small because the magnitude of supply noise, N , is usually small.

Consider now the closed-loop CDAs. On the basis of (14c) and ignoring the first four insignificant terms therein due to the small N , the FBD in the closed-loop CDAs given by (14c) is

$$\text{FBD}_{\text{CL}} \approx \text{FBD}_{\text{CL}}(V_{\text{in}} + V_c) + \text{FBD}_{\text{CL}}(V_{\text{in}} + V_c + V_n) \quad (21)$$

where

$$\begin{aligned} & \text{FBD}_{\text{CL}}(V_{\text{in}} + V_c) \\ &= 20 \log \left(\frac{1}{M} \sum 2J_{\text{CL}}(m\omega_c + k_1\omega_s) \right. \\ & \quad \left. \times (B_1 \cos k_3\pi + B_2 \sin k_3\pi) \right) \text{ dB} \\ & \text{FBD}_{\text{CL}}(V_{\text{in}} + V_c + V_n) \\ &= 20 \log \left(\frac{1}{M} \sum 2J_{\text{CL}}(m\omega_c + k_1\omega_s + k_2\omega_n) \right. \\ & \quad \left. \times (B_1 \cos k_3\pi + B_2 \sin k_3\pi) \right) \text{ dB} \\ & 2\pi \times 20 \text{ Hz} < |k_1\omega_s + m\omega_c + k_2\omega_n| < 2\pi \times 20 \text{ kHz}. \end{aligned}$$

In (21), the first term comprises the $\text{FBD}(V_{\text{in}} + V_c)$ components, and the last term contains the $\text{FBD}(V_{\text{in}} + V_c + V_n)$ components. As in the case for the open-loop CDA, the three parameters that determine the FBD are as follows: 1) carrier frequency; 2) magnitude and frequency of the input signal; and 3) magnitude and frequency of the supply noise. The effects of parameters 1), 2), and 3) for the closed-loop CDA are the same as that for the open-loop CDA. However, the primary difference in the closed-loop CDA over the open-loop CDA is that both $\text{FBD}(V_{\text{in}} + V_c)$ and $\text{FBD}(V_{\text{in}} + V_c + V_n)$ expressed in (21) are independent of N , thereby resulting in nearly equal significance of $\text{FBD}(V_{\text{in}} + V_c)$ and $\text{FBD}(V_{\text{in}} + V_c + V_n)$. In another words, given that parameters 1), 2), and 3) are the same for both the open- and closed-loop CDAs, the FBD in the closed-loop CDA can be worse than the open-loop CDA due to the contribution of $\text{FBD}(V_{\text{in}} + V_c + V_n)$. This finding is significant because, unlike what is generally observed in linear amplifiers, the negative feedback does not mitigate the FBD. This can be attributed to the fact that the FBD is a nonlinearity related to the carrier and that it is dependent on the nonlinear PWM process. Furthermore, the FBD expressed in (21) is true for all closed-loop CDAs.

IV. MEASUREMENTS AND RESULTS

In this section, the models and analytical derivations for PSRR, IMD, and FBD are verified by comparing the derived expressions [(15), (16), (20), and (21)] against HSPICE simulations and on the basis of experimental measurements on a prototype CDA IC and on other CDA realizations using discrete components.

The microphotograph of the prototype CDA IC embodying the 1FB-CDA is shown in Fig. 6. The 2FB-CDA (Fig. 5) is constructed by adding an external loop to the 1FB-CDA IC using discrete components. The circuit parameters of the two CDAs are tabulated in Table I. Note that R_1 and R_2 are realized internally on the IC of the 1FB-CDA, and the remaining resistors and capacitors are discrete components and are connected externally to the prototype IC. The experimental results are obtained

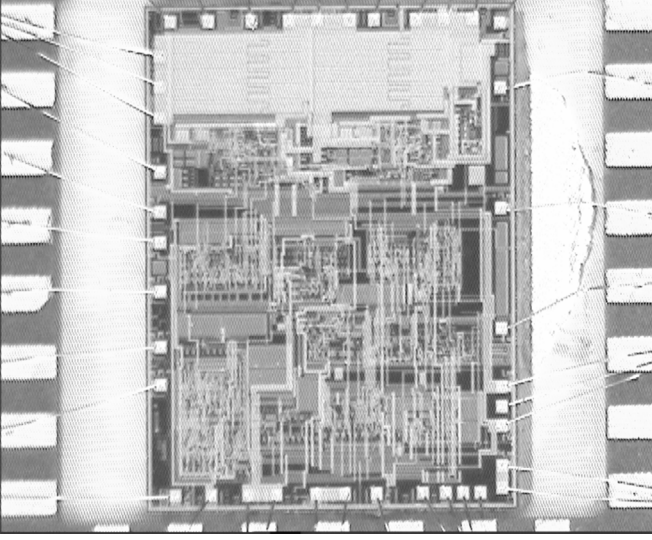


Fig. 6. Microphotograph of a 1FB-CDA.

TABLE I
CIRCUIT PARAMETERS OF THE 1FB- AND 2FB-CDAs

Parameter	Value
V_{DD}	10V
Load	4Ω
$C, R_1, R_2, R_3,$ $R_4, R_5, R_6,$	1nF, 3kΩ, 90kΩ, 20kΩ, 68kΩ, 3kΩ, 90kΩ
Bandwidth of the low-pass filter	>20kHz

by measuring the actual output signal across the load R_L (after the low-pass filter L and C_L).

As the PSRR and IMD for the open-loop CDA are already established [11], no verification is necessary. Fig. 7(a) and (b) shows the PSRR for the 1FB-CDA for the following two conditions: Fig. 7(a), Condition A: $f_n = 100$ Hz, the typical low frequency supply noise, and Fig. 7(b), Condition B: $f_n = 1$ kHz, where the supply noise is relatively high frequency found in typical mixed-signal environments. The LG of the 1FB-CDA design is shown in Fig. 8 and is also expressed in (14e). In Fig. 7 (and in Fig. 10 later), the maximum $M = 0.6$, and this is due to the parameters tabulated in Table I and chosen for the design of the 1FB- and 2FB-CDAs. On the basis of Fig. 7(a) and (b), the following observations are made.

- 1) The analytically obtained PSRR (15) agrees well with that obtained by HSPICE simulations and on the basis of practical measurements, hence verifying the derived analytical expressions.
- 2) As expected, the PSRR is independent of the modulation index M .
- 3) Fig. 7(a) shows that, as expected, the first-harmonic PSRR is insignificant compared with the fundamental PSRR when the supply noise is low frequency ($f_n = 100$ Hz).
- 4) In contrast, Fig. 7(b) shows that the first-harmonic PSRR is significant—only ~ 10 dB higher than (equivalent to $\sim 30\%$ of) the fundamental PSRR—when the supply noise is relatively high frequency ($f_n = 1$ kHz). In short, contrary to

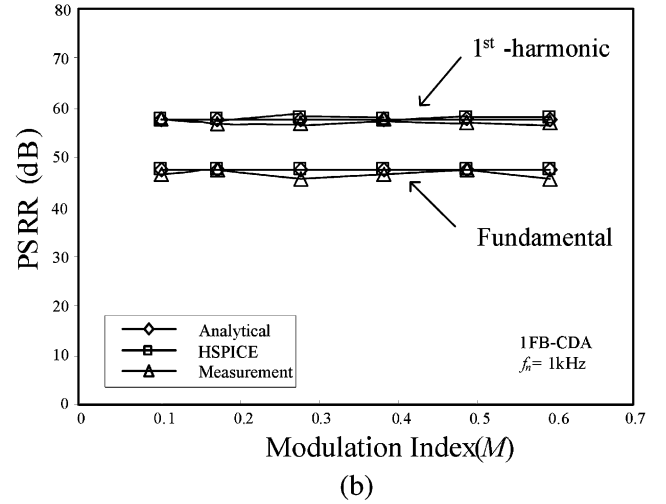
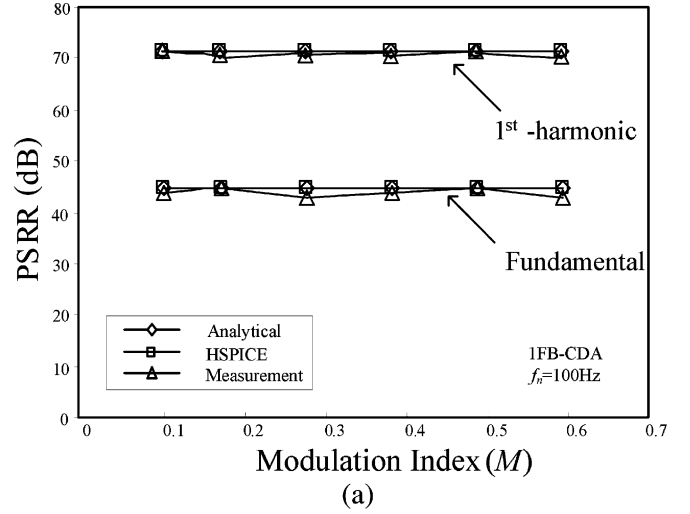
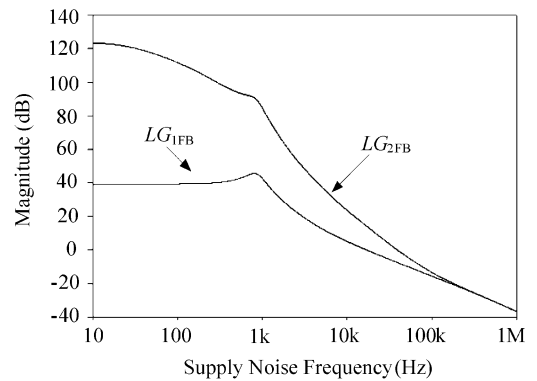
Fig. 7. PSRR (fundamental and first harmonic) of 1FB-CDA. (a) Condition A: $f_n = 100$ Hz. (b) Condition B: $f_n = 1$ kHz.

Fig. 8. Loop gain of the 1FB-CDA and 2FB-CDA.

linear amplifiers, the first-harmonic PSRR components can be significant in closed-loop CDAs in certain conditions.

Fig. 9 shows a comparison of the PSRR of a 1FB-CDA and a 2FB-CDA, where their loop gains are shown in Fig. 8 and where only the fundamental PSRR component is considered. On the basis of Fig. 9, the following observations are made.

- 1) The PSRRs for both the 1FB-CDA and 2FB-CDA topologies obtained analytically agree well with that obtained by

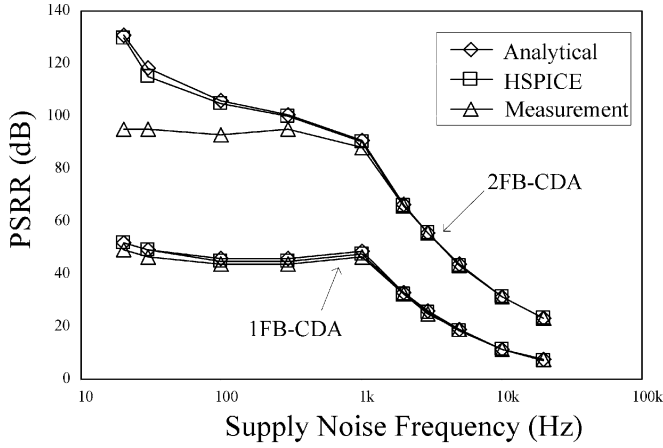


Fig. 9. Comparison of PSRR (fundamental component only) between the 1FB- and 2FB-CDA.

HSPICE simulation and on the basis of practical measurements, hence verifying the derived analytical expressions. The PSRR results also agree with that obtained by the linear model approach [11], hence further verifying the derived analytical expressions (based on a multidimensional Fourier series approach) herein.

- 2) Following 1), the discrepancy in the PSRR in the low-frequency range for the 2FB-CDA topology based on practical measurements is largely due to the floor noise therein.
- 3) The PSRR of the 2FB-CDA at low frequency is significantly better (theoretically ~ 60 -dB improvement at 100 Hz) than the 1FB-CDA, and this is obtained by means of the additional loop filter (increased loop gain given in (15) and Fig. 8). Nonetheless, the degree of improvement decreases as the supply noise frequency increases. This is because, as shown in Fig. 8, the loop gain advantage of the 2FB-CDA over the 1FB-CDA reduces as the frequency increases. The advantages of the 2FB-CDA over the 1FB-CDA are worthwhile despite its added hardware overheads.

Fig. 10(a) and (b) shows the IMD for the 1FB-CDA, and Fig. 10(c) shows the IMD for the 2FB-CDA for the following two conditions: Fig. 10(a), Condition A: $f_n = 100$ Hz, $f_s = 11$ kHz, and $N = 0.1$, and Fig. 10(b) and (c), Condition B: $f_n = 10$ kHz, $f_s = 11$ kHz, and $N = 0.1$. Condition A is the typical low-frequency supply noise, while Condition B is where the supply noise is relatively high frequency. The following observations are made.

- 1) In Fig. 10(a)–(c), the IMD for the 1FB-CDA and 2FB-CDA (16) obtained analytically agrees well with that obtained by HSPICE simulations and on the basis of practical measurements, hence verifying the derived analytical expressions. In Fig. 10(c), the discrepancy in the second-order IMD for the 2FB-CDA is due to the floor noise in practical measurements.
- 2) From Fig. 10(a)–(c), as expected, the IMD in the 1FB- and 2FB-CDA increases when M increases, except for the measured second-order IMD in Fig. 10(c) that is masked due to the floor noise.
- 3) From Fig. 10(a) with Condition A where the supply noise is low frequency ($f_n = 100$ Hz), the third-order IMD in the

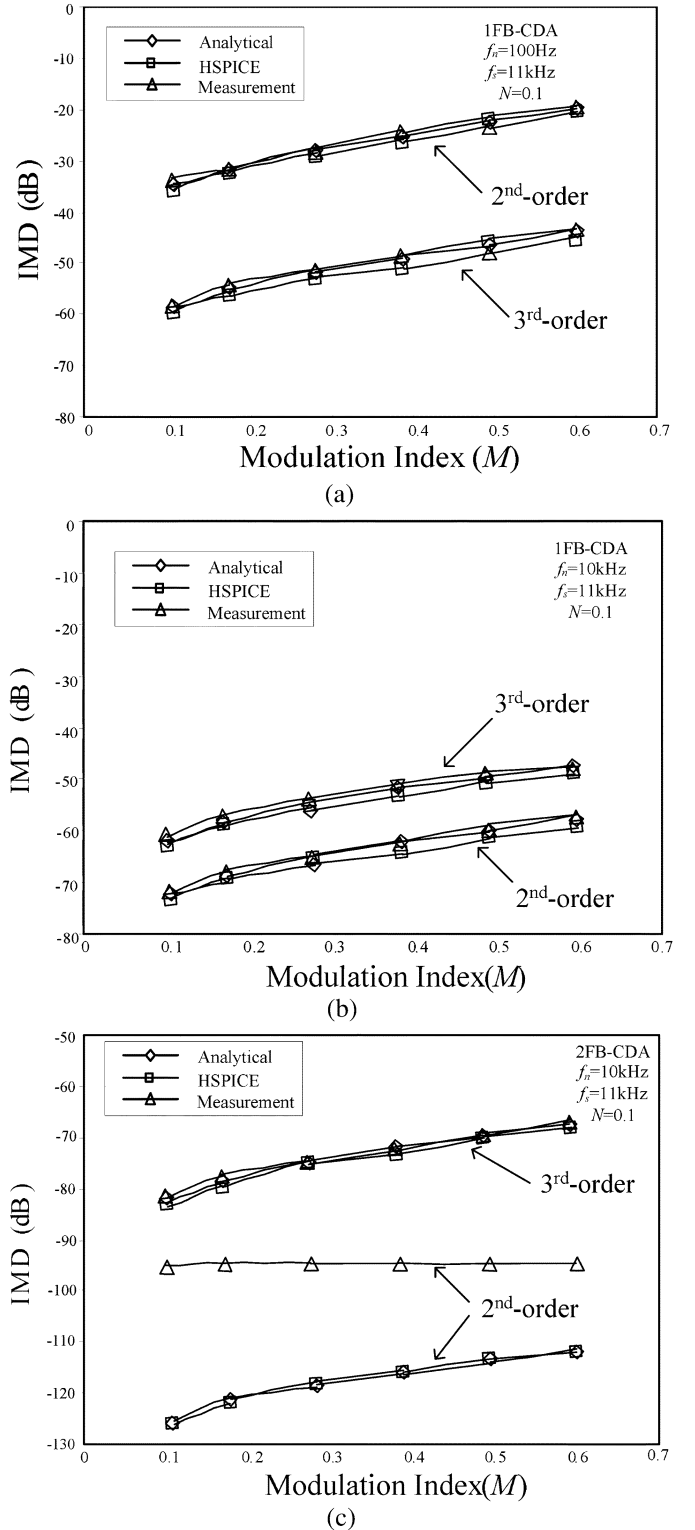


Fig. 10. IMD (second and third order). (a) 1FB-CDA with Condition A: $f_n = 100$ Hz and $f_s = 11$ kHz. (b) 1FB-CDA with Condition B: $f_n = 10$ kHz and $f_s = 11$ kHz. (c) 2FB-CDA with Condition B: $f_n = 10$ kHz and $f_s = 11$ kHz.

1FB-CDA is insignificant compared with the second-order IMD. Although the IMD for the 2FB-CDA with Condition A is not shown, its relationship between second-order (~ -35 dB) and third-order IMDs (~ -62 dB) is similar to that in the 1FB-CDA. The improvement in the 2FB-CDA is due to the increased loop gain therein.

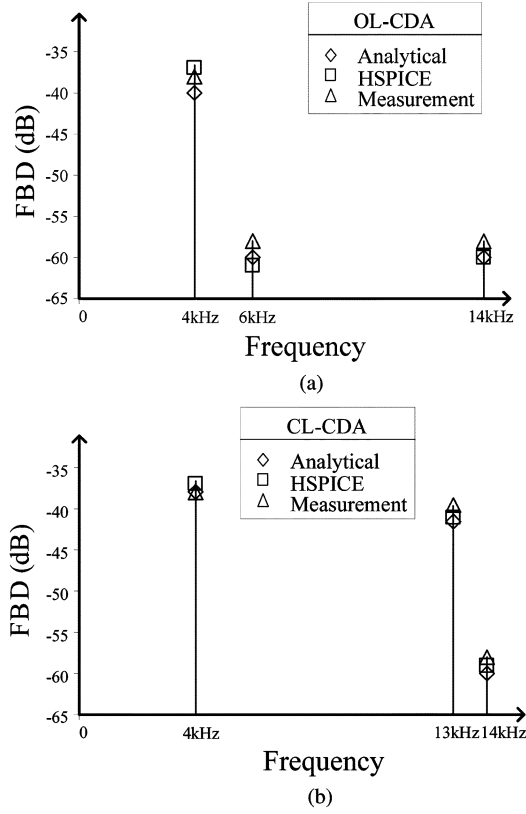


Fig. 11. Significant FBD components in (a) open-loop CDAs and (b) closed-loop CDAs.

- 4) Following 3), however, in Fig. 10(b) with Condition B where the supply noise is high frequency ($f_n = 10$ kHz), the third-order IMD in the 1FB-CDA is more significant (> 10 dB worse) than the second-order IMD. As delineated previously, this result contradicts the one that is typically observed in linear amplifiers.
- 5) Similar to the observations for the 1FB-CDA in 4), the third-order IMD in the 2FB-CDA is also more significant than the second-order IMD.

Fig. 11(a) and (b) shows the FBD for the open- and closed-loop CDAs, respectively, where $f_c = 80$ kHz, $f_s = 19$ kHz, $f_n = 10$ kHz, $M = 0.9$, and $N = 0.1$. The following observations are made.

- 1) The FBD for both the open-loop CDA (20) and the closed-loop CDA (21) obtained analytically agrees well with that obtained by HSPICE simulation and on the basis of practical measurements, hence verifying the derived analytical expressions.
- 2) Fig. 11(a) shows that the FBD components of the open-loop CDA are at 4, 6, and 14 kHz, where the $FBD(V_{in} + V_c)$ component is at 4 kHz and the $FBD(V_{in} + V_c + V_n)$ components due to the supply noise are at 6 and 14 kHz. Among the three components, the $FBD(V_{in} + V_c)$ component at 4 kHz is the most significant, and the $FBD(V_{in} + V_c + V_n)$ components at 6 and 14 kHz are insignificant due to the small N .
- 3) Fig. 11(b) shows that the FBD components of the closed-loop CDA are at 4, 13, and 14 kHz, where the $FBD(V_{in} +$

$V_c)$ component is at 4 kHz and the $FBD(V_{in} + V_c + V_n)$ components are at 13 and 14 kHz. Among the three components, both the $FBD(V_{in} + V_c)$ component at 4 kHz and the $FBD(V_{in} + V_c + V_n)$ component at 13 kHz are equally significant. In other words, the now-defined $FBD(V_{in} + V_c + V_n)$ is a significant component of the overall FBD.

- 4) On the basis of 2) and 3), it is interesting to observe that the FBD in the closed-loop CDA is worse than that in the open-loop CDA. As delineated earlier, this observation is unlike what is generally observed in linear amplifiers pertaining to the benefits of negative feedback on nonlinearities. For completeness, note that if f_c is increased, for example, from 80 to 250 kHz, the FBD components are considerably attenuated to below the floor noise.

In summary, the analytical expressions have been verified, and they provide an analytical means to quantify PSRR, IMD, and FBD, and provide insight into the mechanisms therein and how and what parameters affect the nonlinearities. These results also show that the first-harmonic PSRR and third-order IMD components are significant under certain conditions and that the negative-feedback closed-loop CDAs may be disadvantageous over the rudimentary open-loop CDAs in terms of FBD.

V. CONCLUSION

The application of Fourier series analyses on prevalent CDA designs, namely, open-loop, 1FB-, and 2FB-CDAs, has been presented, where all nonlinear components due to the noisy supply have been modeled. The analytical expressions of PSRR, IMD, and FBD therein have been derived. It has been shown that unlike linear amplifiers, the first-harmonic PSRR and third-order IMD components of CDAs could be significant compared with the fundamental PSRR and second-order IMD components, respectively, and that high PSRR and low IMD can be achieved by increasing the loop gain. It has also been shown that the overall FBD can increase due to the supply noise and that the FBD in closed-loop CDAs, due to feedback, can be worse than that in open-loop CDAs. The analyses have been verified by comparing the analytical results against HSPICE simulations and on the basis of practical measurements on a prototype CDA IC and on other discrete realizations.

APPENDIX

DERIVATIONS OF THE PWM SIGNAL WITH NONLINEARITIES DUE TO THE SUPPLY NOISE

To derive (16b), first, substitute (13) into (14a)

$$V_{1FB+N}(t) = 1 - [1 - \zeta_n(t)](1 + N \cos \omega_n t) + (1 + N \cos \omega_n t) \sum_{m=1}^{\infty} \frac{2}{m\pi} \sin(m\pi \zeta_n) \cos(m\omega_c t). \quad (A1)$$

By substituting (12) into (A1), the first two terms of (14b) are easily obtained

$$V_{1FB+N}(t) = \frac{1}{2} + \frac{M}{2} \cos \omega_s t + V_{1FB+N}(t)_3 - V_c. \quad (A2)$$

The derivation for the third and fourth terms of (A2) will now be presented in turn.

A. Derivation of the Third Term $V_{1\text{FB}+N}(t)_3$

$V_{1\text{FB}+N}(t)_3$ is expressed as

$$V_{1\text{FB}+N}(t)_3 = (1 + N \cos \omega_n t) \times V_\Sigma \quad (\text{A3})$$

where

$$V_\Sigma = \sum_{m=1}^{\infty} \frac{2}{m\pi} \sin(m\pi \zeta_n) \cos(m\omega_c t). \quad (\text{A4})$$

The duty cycle can be decomposed as

$$\begin{aligned} \zeta_n(t) &= \frac{1}{2} + \frac{1}{2} M \cos \omega_s t \sum_{i=0}^{\infty} (-1)^i N^i \cos^i \omega_n t \\ &\quad + \frac{1}{2} \sum_{j=0}^{\infty} (-1)^j N^{j+1} \cos^{j+1} \omega_n t \\ &\quad - V_e \sum_{k=0}^{\infty} (-1)^k N^k \cos^k \omega_n t \end{aligned} \quad (\text{A5})$$

$$\begin{aligned} &\approx \frac{1}{2} + \frac{1}{2} M \cos \omega_s t + \frac{1}{2} N \cos \omega_n t \\ &\quad - \frac{1}{2} MN \cos \omega_s t \cos \omega_n t. \end{aligned} \quad (\text{A6})$$

Equation (A6) is an approximated expression of the duty cycle $\zeta_n(t)$, and the remaining terms in (A5) are inconsequential, so they are ignored. Note that all the terms related to V_e are negligible due to the small magnitude of V_e .

Substituting (A6) into (A4) as

$$\begin{aligned} V_\Sigma &= \sum_{m=1}^{\infty} \frac{2}{m\pi} \Re \left(e^{j\frac{m\pi}{2}} e^{j\frac{Mm\pi \cos \omega_s t}{2}} e^{j\frac{Nm\pi \cos \omega_n t}{2}} \right. \\ &\quad \left. \times e^{-j\frac{MNm\pi \cos \omega_s t \cos \omega_n t}{2}} \right) \cos(m\omega_c t) \end{aligned} \quad (\text{A7})$$

where $\Re[\cdot]$ is the operator that takes only the imaginary part.

Using the following mathematical relationship: as

$$e^{jm\gamma \sin \theta} = \sum_{q=-\infty}^{\infty} J_q(m\gamma) e^{jq\theta}. \quad (\text{A8})$$

Equation (A7) can be expressed

$$\begin{aligned} V_\Sigma &= \sum_{m=1}^{\infty} \frac{2 \cos(m\omega_c t)}{m\pi} \sum_{a=-\infty}^{\infty} \sum_{b=-\infty}^{\infty} \sum_{c=-\infty}^{\infty} \sum_{d=-\infty}^{\infty} \\ &\quad \times \left\{ J_a \left(\frac{Mm\pi}{2} \right) J_b \left(\frac{Nm\pi}{2} \right) J_c \left(\frac{MNm\pi}{4} \right) J_d \left(\frac{MNm\pi}{4} \right) \right. \\ &\quad \left. \times \Re \left\{ e^{j \left[\frac{(a+b+m-c-d)\pi}{2} + (-a+c+d)\omega_s t + (-b-c+d)\omega_n t \right]} \right\} \right\} \\ &= \sum_{m=1}^{\infty} \sum_{a=-\infty}^{\infty} \sum_{b=-\infty}^{\infty} \sum_{c=-\infty}^{\infty} \sum_{d=-\infty}^{\infty} \\ &\quad \times J_z [2B_1 \cos k_3 \pi \sin(k_1 \omega_s + k_2 \omega_n + m\omega_c) t \\ &\quad + 2B_2 \cos k_3 \pi \cos(k_1 \omega_s + k_2 \omega_n + m\omega_c) t] \end{aligned} \quad (\text{A9})$$

By substituting (A9) into (A3), $V_{1\text{FB}+N}(t)_3$ is obtained as

$$\begin{aligned} V_{1\text{FB}+N}(t)_3 &= \sum_{m=1}^{\infty} \sum_{a=-\infty}^{\infty} \sum_{b=-\infty}^{\infty} \sum_{c=-\infty}^{\infty} \sum_{d=-\infty}^{\infty} J_z \\ &\quad \times \{ B_1 N \cos(k_3 \pi) \sin[k_1 \omega_s + (k_2 + 1)\omega_n + m\omega_c] t \\ &\quad + B_1 N \cos(k_3 \pi) \sin[k_1 \omega_s + (k_2 - 1)\omega_n + m\omega_c] t \\ &\quad + B_2 N \sin(k_3 \pi) \cos[k_1 \omega_s + (k_2 - 1)\omega_n + m\omega_c] t \\ &\quad + B_2 N \sin(k_3 \pi) \cos[k_1 \omega_s + (k_2 + 1)\omega_n + m\omega_c] t \\ &\quad + 2B_1 \cos(k_3 \pi) \sin(k_1 \omega_s + k_2 \omega_n + m\omega_c) t \\ &\quad + 2B_2 \sin(k_3 \pi) \cos(k_1 \omega_s + k_2 \omega_n + m\omega_c) t \} \end{aligned} \quad (\text{A10})$$

where

$$\begin{aligned} B_1 &= [1 - (-1)^{k_1+k_2}] / 4 & B_2 &= [1 + (-1)^{k_1+k_2}] / 4 \\ k_1 &= -a + c + d & k_2 &= -b - c + d \\ k_3 &= (a + b + m - c - d) / 2 \\ J_z &= \frac{1}{m\pi} J_a \left(\frac{mM\pi}{2} \right) J_b \left(\frac{mN\pi}{2} \right) \\ &\quad \times J_c \left(\frac{mMN\pi}{4} \right) J_d \left(\frac{mMN\pi}{4} \right). \end{aligned}$$

B. Derivation of the Fourth Term V_e

Since V_e comprises all the nonlinearities relating to PSRR and IMD, by considering the PSRR and IMD derived from the linear model [11], V_e can be expressed as

$$V_e \approx \frac{N \cos \omega_n t}{2(\text{LG}(\omega_n) + 1)} + \frac{MN \cos(\omega_s \pm \omega_n) t}{4(\text{LG}(\omega_s \pm \omega_n) + 1)} + \varepsilon \quad (\text{A11})$$

where ε comprises the remaining PSRR and IMD components.

Consider now the derivation of ε in (A11), produced by the intermodulation between $V_{1\text{FB}}$ and the supply noise. On the basis of the first and second terms in (A11), the components in $V_{1\text{FB}}$ that are related to PSRR and IMD can be expressed as

$$\begin{aligned} V_{1\text{FB}_e} &= -\frac{\text{LG}(\omega_n) N \cos \omega_n t}{2(\text{LG}(\omega_n) + 1)} \\ &\quad - \frac{\text{LG}(\omega_s \pm \omega_n) MN \cos(\omega_s \pm \omega_n) t}{4(\text{LG}(\omega_s \pm \omega_n) + 1)}. \end{aligned} \quad (\text{A12})$$

The components expressed in (A12) further intermodulate with the injected supply noise at the output stage, and the resulting nonlinearities are suppressed by the associated loop gain. Following the derivations of the fundamental PSRR and second-order IMD components in the open-loop CDA (6) and the 1FB-CDA (A11), the remaining PSRR and IMD components in V_e can be derived as

$$\varepsilon = \frac{N^2 \cos 2\omega_n t}{4(\text{LG}(2\omega_n) + 1)} + \frac{MN^2 \cos(\omega_s \pm 2\omega_n) t}{8(\text{LG}(\omega_s \pm 2\omega_n) + 1)} + \dots \quad (\text{A13})$$

Following the same procedures for (A12) and (A13), other harmonic PSRR components and higher order IMD components can be derived, and as these components can be shown to be insignificant, they are ignored.

Finally, by substituting (A13) into (A11), V_e is obtained.

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